# STAFF REVIEW · H446 027 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

## Do Now — source answers

1. $0.1100 \times 2^3$ . Shift the mantissa one place left and decrease the exponent by one to preserve the value. A normalised positive two’s-complement mantissa begins 01 ; a negative one begins 10 .

2. Indexes 3 → 5 → 4 (values 19 → 35 → 27). Binary search tests the middle index of the remaining sorted range, using zero-based indexing.

3. [3, 5, 2, 7] . Compare adjacent values and swap when the left value is larger. The largest value, 7, moves to the end.

4. $Q = (A \wedge B) \vee C$ . A = door open, B = armed, C = panic pressed. Group door AND armed first; panic is an alternative trigger.

## I Do — source worked steps

1. Outer NOT over AND — CODE:
Use De Morgan's Law.

2. Negate each term — CODE:
$A \rightarrow \neg A$
$B \rightarrow \neg B$

3. AND becomes OR — CODE:
$\neg A \vee \neg B$

4. Equivalent behaviour — CODE:
Both are false only when A=1 and B=1.

## We Do — guided CHECK

1. Complete the same expression
CODE:
$\neg A \vee \neg B$
$= \neg(A \wedge B)$

2. Complete the same expression
CODE:
$\neg(\neg B)$
$= ?$
CODE:
$\neg(\neg B)$
$= B$

3. Complete the same expression
CODE:
$(A \vee B) \wedge (A \vee C)$

= A ∨ (? ? ?)
CODE:
(A ∨ B) ∧ (A ∨ C)
= A ∨ (B ∧ C)

4. Complete the same expression
CODE:
C ∨ (A ∨ B)
= ?
CODE:
C ∨ (A ∨ B)
= (A ∨ B) ∨ C

## You Do — source mark schemes

1. (i) Double negation
CODE:
¬¬A = A
Two NOT operations cancel.
(ii) De Morgan’s law
CODE:
¬A ∧ ¬B = ¬(A ∨ B)
NOT A AND NOT B means neither A nor B is true. This is an equivalent expression, not necessarily a shorter one.
(iii) De Morgan’s law, then double negation
CODE:
¬(¬A ∧ ¬B)
= ¬¬A ∨ ¬¬B
= A ∨ B

2. Step 1 — identify the repeated factor E.
CODE:
(E ∧ F) ∨ (E ∧ G)
Step 2 — use the distributive law.
CODE:
= E ∧ (F ∨ G)
Read it back: the voucher must be shown AND either a soft drink OR water must be purchased.

3. Step 1 — under-18 route: under 18 AND student card.
CODE:
A ∧ B
Step 2 — over-60 route: over 60 AND ID.
CODE:
C ∧ D
Step 3 — either route qualifies, so combine with OR.
CODE:
Q = (A ∧ B) ∨ (C ∧ D)

4. Step 1 — first OR gate:
CODE:

X = A ∨ B
Step 2 — second OR gate:
CODE:
Y = C ∨ D
Step 3 — final OR gate joins X and Y:
CODE:
Q = X ∨ Y
= (A ∨ B) ∨ (C ∨ D)
Because OR is associative, this also equals A ∨ B ∨ C ∨ D.

## Exit Ticket — source answer

1. True. De Morgan's laws complement each input and exchange AND with OR when negating an expression.